Lesson 7.2 Guided Notes

Session 1 · Rational Numbers on a Number Line

I can plot a rational number and its opposite on a number line. · 6.NS.C.6c

Word / Spanish	Meaning — complete the blank	Example
rational number / número racional	A number that can be written as a _____ of two integers.	−3/4, 0.5, 7 = 7/1
integer / número entero	A whole number or its _____.	−3, 0, 5
number line / recta numérica	A line where numbers _____ from left to right.	−2 is left of 1, so −2 < 1

Key idea

To plot 7/4, split each whole into _____ equal parts and count _____ parts from 0.

Negative numbers sit to the _____ of 0 on a horizontal line and _____ 0 on a vertical line.

Worked example

Plot −2.75 on the number line.

−2.75 is between the integers _____ and _____.

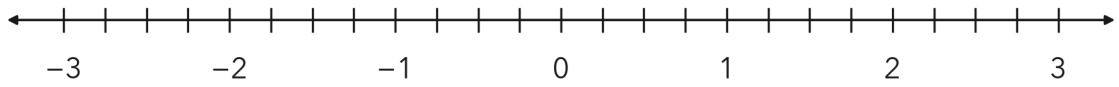
Split the gap between −3 and −2 into _____ equal parts (fourths).

Start at −2 and count _____ fourths to the left.

Answer: −2.75 is one fourth to the _____ of −3.

You try

1. Plot and label −1.5 and 2.25 on the number line.



_____

2. Name the two integers each number is between.
a. −3.6 _____
b. 5/2 _____

_____

Sum it up: To plot a rational number on a number line, I first ___ because ___.

_____

Session 2 · Opposites of Rational Numbers

I can find the opposite of a rational number and explain what taking the opposite twice does. · 6.NS.C.6a

Word / Spanish	Meaning — complete the blank	Example
opposites / opuestos	Two numbers the same _____ from 0, on different sides of 0.	−2.5 and 2.5
opposite of the opposite / opuesto del opuesto	Taking the opposite _____ gives back the number you started with.	−(−3) = 3
zero / cero	The only number that is its _____ opposite.	the opposite of 0 is 0

Key idea

The symbol −(4.5) means the _____ of 4.5, which is −4.5.

The opposite of a negative number is _____.

Worked example

Find −(−5.3).

−5.3 is _____ units to the left of 0.

Its opposite is the same distance on the _____ side of 0.

So −(−5.3) = _____.

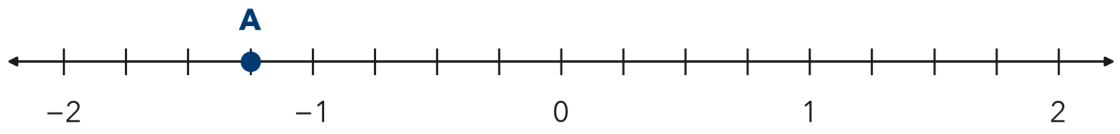
Answer: −(−5.3) = _____

You try

1. Find each value.
a. opposite of 2/9 _____
b. −(−0.6) = _____
c. opposite of 0 _____

_____

2. Point A is at −1.25. Plot its opposite and label it B.



_____

Sum it up: Taking the opposite twice gives ___ because ___.

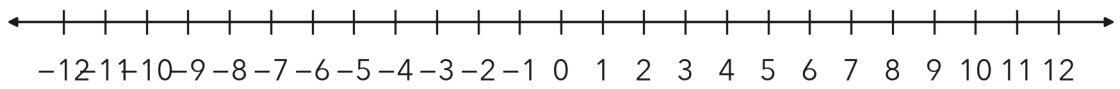
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Session 1 · Model, reason, and explain

Rational Numbers on a Number Line · 6.NS.C.6c

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Plot −1.75 between −2 and −1 on a number line divided into fourths. Explain the value of one interval and the point's position.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: counts ticks without identifying their scale.. Explain what changes and what stays true.

_____

_____

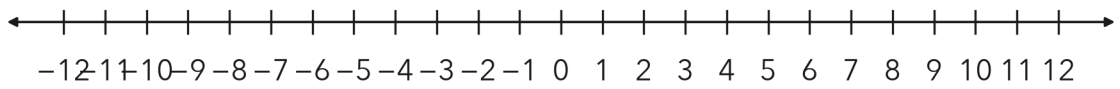
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Opposites of Rational Numbers · 6.NS.C.6a

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Plot −3/2 and its opposite. Rewrite them as decimals and explain opposite of the opposite.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: changes notation but not location or confuses opposite with reciprocal.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?